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Safety gates for elevated platforms, and related methods of use

Abstract

A safety gate has: a structural frame including a pair of columns spaced from one another along a peripheral edge of an elevated platform to define a ledge entrance to a staging area of the elevated platform; a ledge gate; and a staging gate; and in which the ledge gate and the staging gate are connected to move together relative to the pair of columns: into a loading configuration when the ledge gate rises and pitches upward, and the staging gate lowers and unfolds laterally away from the pair of structural columns, to allow a) the ledge gate to open the ledge entrance and b) the staging gate to enclose the staging area; and into an unloading configuration when the ledge gate lowers and pitches downward, and the staging gate rises and folds laterally toward the pair of structural columns, to allow a) the ledge gate to close the ledge entrance and b) the staging gate to expose the staging area.

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Background/Summary

TECHNICAL FIELD

(1) This document relates to safety gates for elevated platforms, and related methods of use.

BACKGROUND

(2) The following paragraphs are not an admission that anything discussed in them is prior art or part of the knowledge of persons skilled in the art.

(3) Safety gates exist for mezzanine use in industrial warehouses, operating via several different mechanisms, including garage-door-style rail and panel designs, cantilever teeter-totter designs, guillotine-style designs, and vertical-moving cage designs.

SUMMARY

(4) A safety gate is disclosed comprising: a structural frame including a pair of columns spaced from one another along a peripheral edge of an elevated platform to define a ledge entrance to a staging area of the elevated platform; a ledge gate; and a staging gate; and in which the ledge gate and the staging gate are connected to move together relative to the pair of columns: into a loading configuration when the ledge gate rises and pitches upward, and the staging gate lowers and unfolds laterally away from the pair of structural columns, to allow a) the ledge gate to open the ledge entrance and b) the staging gate to enclose the staging area; and into an unloading configuration when the ledge gate lowers and pitches downward, and the staging gate rises and folds laterally toward the pair of structural columns, to allow a) the ledge gate to close the ledge entrance and b) the staging gate to expose the staging area.

(5) A method is disclosed comprising: moving a ledge gate and a staging gate, relative to a pair of columns, into a loading configuration, the pair of columns forming a structural frame and being spaced from one another along a peripheral edge of an elevated platform to define a ledge entrance to a staging area of the elevated platform, in which the ledge gate and staging gate are connected to move together, and in which while moving into the loading configuration the ledge gate rises and pitches upward, and the staging gate lowers and unfolds laterally away from the pair of structural columns, to allow a) the ledge gate to open the ledge entrance and b) the staging gate to enclose the staging area; and moving the ledge gate and the staging gate, relative to the pair of columns, into an unloading configuration, in which while moving into the unloading configuration the ledge gate lowers and pitches downward, and the staging gate rises and folds laterally toward the pair of structural columns, to allow a) the ledge gate to close the ledge entrance and b) the staging gate to expose the staging area.

(6) A safety gate is disclosed comprising: a structural frame including a pair of columns spaced from one another along a peripheral edge of an elevated platform to define a ledge entrance to a staging area of the elevated platform; a ledge gate; and a staging gate; in which the ledge gate and the staging gate are connected to move together relative to the pair of columns: into a loading configuration when the ledge gate rises and pitches upward, and the staging gate moves, to allow a) the ledge gate to open the ledge entrance and b) the staging gate to enclose the staging area; and into an unloading configuration when the ledge gate lowers and pitches downward, and the staging gate moves, to allow a) the ledge gate to close the ledge entrance and b) the staging gate to expose the staging area; and in which the ledge gate is mounted to move along opposed rails on the pair of columns, the opposed rails define a curved portion of a path along which the ledge gate pitches during movement, the ledge gate defines a free base end and top end, and is mounted to pitch upward and downward via a first pivot axis intermediate the free base end and the top end.

(7) Embodiments are disclosed where one or more of: a. the gate orientation allows relatively greater clearance for pallet loads without making the frame taller; b. the user has the ability to access a load on the mezzanine from three sides on the horizontal plane and to be picked up by

crane from above (other gates have obstructions that prevent access from two or more directions); c. the gate has a compact design that minimizes the overall footprint when in the closed position; d. stiffeners/guards are integrated on the “front” of the gate (facing off the mezzanine) to limit forklift damage; e. an installation system is provided that allows the gate to be installed on multiple types of mezzanines (concrete, steel, grating, wood etc.); and f. integration points are provided to integrate with existing handrails.

(8) In various embodiments, there may be included any one or more of the following features: The ledge gate is mounted to move along opposed rails on the pair of columns. The opposed rails define a curved portion of a path along which the ledge gate pitches during movement. The curved portion is an upper portion of the path. The opposed rails are structured to define the path with an upside-down-J-shape. The ledge gate defines a free base end and top end, and is mounted to pitch upward and downward via a first pivot axis intermediate the free base end and the top end. The ledge gate is mounted to pitch upward and downward via a second pivot axis intermediate the first pivot axis and the top end. The ledge gate is mounted to pitch about a pitch axis perpendicular to the pair of columns and parallel with the peripheral edge. A cable secured between the staging gate and the ledge gate, and supported on a cable guide mounted to the pair of columns above the ledge gate. The cable is secured to a free base end of the ledge gate. The cable comprises one or more of a rope, a chain, a belt, or a strap. The cable guide comprises one or more of a pulley, sprocket, bushing, or a sheave. The cable comprises or connects to a biasing device. The biasing device comprises a stretch limiter. The biasing device connects the cable to one or both of the staging gate and the ledge gate. The ledge gate is connected to move: into the loading configuration when a free base end of the ledge gate pitches outward laterally away from the pair of columns and the staging area; and into the unloading configuration when the free base end pitches inward laterally toward the pair of columns and the staging area. The ledge gate is oriented vertically in the unloading configuration, and horizontal or near-horizontal in the loading configuration. The ledge gate is oriented vertically in the unloading configuration, and approximately forty five degrees in the loading configuration. The staging gate comprises an end gate wall and opposed side gate walls. The end gate wall, opposed side gate walls, and the ledge cooperate in the loading configuration to define the staging area as having a rectangular shape. The opposed side gate walls are pivotally attached to the pair of columns and the end gate wall. The opposed side gate walls form respective four bar mechanical linkages that fold and deploy between the unloading and loading configurations, respectively. The elevated work platform is a mezzanine in a building. While the ledge gate and staging gate are in the loading configuration, depositing a load into the staging area via the ledge entrance. Depositing is done using a fork lift, and the load comprises a pallet. While the ledge gate and the staging gate are in the unloading configuration, removing a load from the staging area. Removing the load is done using a fork lift, and the load comprises a pallet. Removing the load is done using a crane.

(9) The foregoing summary is not intended to summarize each potential embodiment or every aspect of the subject matter of the present disclosure. These and other aspects of the device and method are set out in the claims.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) Embodiments will now be described with reference to the figures, in which like reference characters denote like elements, by way of example, and in which:

(2) FIG. 1 is a front perspective view of a safety gate in a loading configuration, with a mezzanine ledge, mezzanine railing, and staging area shown in dashed lines.

(3) FIG. 2 is a front perspective view of another safety gate in a loading configuration, with a

mezzanine ledge and staging area shown in dashed lines.

(4) FIG. 3 is a side elevation view of the safety gate of FIG. 2 in the loading configuration with the mezzanine ledge, staging area, pallet load, and forklift tines shown in dashed lines.

(5) FIG. 4 is a rear perspective view of the safety gate of FIG. 2 in the loading configuration with the mezzanine ledge and staging area shown in dashed lines.

(6) FIG. 5 is a top plan view of the safety gate of FIG. 2 in the loading configuration with the mezzanine ledge, staging area, pallet load, and forklift tines shown in dashed lines.

(7) FIG. 6 is a rear elevation view of the safety gate of FIG. 2 in the loading configuration with the mezzanine ledge, staging area, and pallet load shown in dashed lines.

(8) FIG. 7 is a front perspective view of the safety gate of FIG. 2 in an unloading configuration, with the mezzanine ledge shown in dashed lines.

(9) FIG. 8 is a side elevation view of the safety gate of FIG. 7 in the unloading configuration with the mezzanine ledge, staging area, and pallet load shown in dashed lines, and a crane hook shown in position to lift the pallet load.

(10) FIG. 9 is a rear perspective view of the safety gate of FIG. 7 in the unloading configuration with the mezzanine ledge and staging area shown in dashed lines.

(11) FIG. 10 is a top plan view of the safety gate of FIG. 7 in the unloading configuration with the mezzanine ledge, staging area, and pallet load, shown in dashed lines.

(12) FIG. 11 is a front elevation view of the safety gate of FIG. 7 in the unloading configuration with the mezzanine ledge, staging area, and pallet load shown in dashed lines.

(13) FIG. 12 is a front perspective view of another embodiment of a safety gate, illustrated in a loading configuration, with a mezzanine ledge shown in dashed lines.

(14) FIG. 12A is a close up view of an area denoted by dashed lines in FIG. 12.

(15) FIG. 13 is a side elevation view of the safety gate of FIG. 12 in the loading configuration with the mezzanine ledge and staging area shown in dashed lines.

(16) FIG. 14 is a side elevation view of the safety gate of FIG. 12 in an unloading configuration with the mezzanine ledge and staging area shown in dashed lines.

DETAILED DESCRIPTION

(17) Immaterial modifications may be made to the embodiments described here without departing from what is covered by the claims.

(18) In buildings with high ceilings such as industrial warehouses, fulfillment centers, record retention centers, factories, chemical plants, and retail stores, a raised platform, also known as a mezzanine, may be used to optimize the storage density of equipment and materials and/or to increase the amount and utilization of floor space. These work platforms may be permanent or nonpermanent, engineered, structures. A mezzanine may be configured to be constructed and deconstructed separately from the building in which it is housed, thus allowing the facility to be flexibly repurposed. A mezzanine may include staircases, walkways, and elevated work areas accessible by personnel. A mezzanine may be used for overhead access of equipment, for example to facilitate the access of the top of process equipment. In one example warehouse personnel may use an elevated platform to empty a feedstock into a hopper at the top of blender or bagger machine, or to load and unload parts into a chemical bath. In the most common scenario, a mezzanine is used for storage, for example of various loads supported on wooden or other types of pallets. Elevated work platforms may be designed to permit access with forklifts, lift trucks, pallet lifters, and overhead cranes for the transfer of heavy loads.

(19) Various types of machinery may be used to transport, lift, and lower a pallet within a warehouse. A pallet jack, also known as a pallet truck, pallet pump, pump truck, scooter, dog, or jigger is a tool used to lift and move pallets. Pallet jacks are the most basic form of a forklift and are intended to move pallets within a warehouse. A pallet jack is a mechanism that may be leveraged via manual or powered means. A classic example of a manually operated pallet jack is the tool most commonly referred to as a pallet jacket, and may incorporate a pneumatic or

hydraulic mechanism that a user may crank to lift a pallet. A classic powered machine that operates a pallet jack is a forklift. A forklift (also called a lift truck, jitney, fork truck, fork hoist, and forklift truck) is a powered industrial truck used to lift and move materials over short distances. Forklifts are a critical element of any warehouse and distribution center. A forklift may be available in many variations and load capacities. The forklift is designed with a load limit for the forks (also called tines) which is decreased with fork elevation and undercutting of the load (i.e., when a load does not butt against the fork “L”). In a typical warehouse setting, most forklifts have load capacities between one and five tons, while larger machines, for example having up to 50 tons lift capacity, may be used for lifting heavier loads, such as loaded shipping containers. In addition to a control to raise and lower the forks (also known as blades or tines), the operator may be able to tilt the mast backward to compensate for a load's tendency to angle the blades toward the ground and risk slipping off the forks. Tilt also provides a limited ability to operate on non-level ground.

(20) An elevated work platform in a building may incorporate various types of infrastructure to facilitate the storage and access of pallet-supported loads. Pallet racking is a type of elevated work platform common within industrial warehouses. Pallet racks are commonly used to store plural pallets of goods. The general structure of a pallet rack may be exploited to contain other items such as a picking module. A pick module may be present in the form of multi-level racking structures with conveyors throughout, in order to move products from the picking areas to the shipping areas. In the simplest example of a mezzanine, the elevated work platform comprises a level platform whose area may be used for storage, work, or other purposes.

(21) Another example of an elevated work platform is a loading dock of a building. Industrial and other facilities may be equipped with loading docks that allow equipment to be moved in and out of tractor-trailer cargo bays or rail cars without having to lift or lower the equipment. The loading dock may be positioned at the same height as a tractor-trailer cargo bay so that personnel and forklifts may access the cargo bay without raising or lowering the materials (i.e. the load). When a tractor-trailer is not at the loading dock, the loading dock may form a ledge that presents a fall hazard for personnel, and even with the tractor-trailer is present there may be a hazardous gap or partial ledge. The loading dock may thus have need for safety apparatus at all times.

(22) Referring to FIG. 1, an elevated work platform **12** may have railings or gates to mitigate the risk of falls to workers. Throughout industry, a mezzanine may be required by law (for example, occupational health and safety (OH&S) regulations derived from the Canada Labour Code) to encircle raised platforms with railings **48** (often, with a railing at least forty two inches high, a railing at a lower height and a toe board at the level of the platform). An elevated work platform **12** may present a serious fall hazard, which makes safety railings **48** critical to reduce or eliminate hazards of working at height. Safety railings **48** may have various suitable structures, such as laterally spaced columns **48B** supporting horizontal beams **48A**. Other structures may be used, including mesh or solid fencing.

(23) A safety risk arises when a section of a ledge of an elevated work platform must be opened in order to deposit or remove a load to or from the elevated work platform. The presence of safety railings inhibits the ability of a forklift to raise material to the mezzanine. To meet the requirement for worker safety while allowing the movement of pallets on and off a mezzanine, railing may be provided with safety gates that can be temporarily opened for forklift access. A simple type of a safety gate is a sliding or pivoting section of railing that can be moved aside to open a gap in a railing. This style of gate allows forklift access and does not occupy much space, either on the work platform or in the space outside the work platform where the forklift may approach it. However, it does not eliminate the risk of fall, since, while the gate is open, there is no barrier between the ledge and a worker when a forklift is loading on or unloading a pallet from the platform. In order to address the inherent risk in even a simple, gate, various improvements of the basic safety gate have been attempted.

(24) Some safety gates may incorporate two interlocked gates, one termed a ledge gate and another

termed a rear gate. The ledge gate may form a segment of the railing when a forklift is not loading or unloading a pallet onto an elevated platform. The rear gate or staging gate may form a barrier between workers and a demarcated staging area from which a loaded pallet can be added or removed by a forklift. The gates may be interlocked such that both cannot be open or closed simultaneously, so that there is always a barrier between personnel and ledge. A safety gate may be manually operated or operated with the help of a powered actuator, for example an electric motor or hydraulic actuator. If the two gates are physically interlocked, the gates may be counterbalanced so that opening one tends to facilitate the closing of the other due to the shift of weight within the mechanism.

(25) One type of a safety gate includes a pivoting structure, similar to a teeter totter, for example as disclosed in U.S. Pat. No. 5,709,050. This type of safety gate may have a rigid cage that pivots about a fixed fulcrum mounted on a section of or extension to a railing. The rigid cage may have two gates mounted at right angles to each other, enclosing three sides of the staging area and only one of which is pivoted into place at a time. This style of safety gate occupies a relatively large envelope. For example, if the staging area is a cube of equal dimensions, a safety gate of this construction may fill an envelope twice as high as the staging area and may swing into the aisle space as it shifts from its loading position to its closed position. Furthermore, when in the closed position, such a design may limit access to a pallet from all but one side and may fill a lateral space as wide as the staging area, even when in the closed position. It may also prevent access to a pallet from above when in the closed position, which may prevent unloading it through the use of an overhead crane.

(26) Another type of safety gate may include a vertically translating front or rear gate that may be raised and lowered by a cable, rope, or belt, for example as disclosed in U.S. Pat. No. 5,592,779. Such a style of gate may operate similar to a guillotine. The front and rear gates may be substantially planar frames, or may be three-sided fences that delineate the staging area when in the loading position. The front and rear gates may be coupled using a cable such that only one of the two gates may be open at the same time. In some constructions, a rigid linkage may be used in lieu of a cable so that when one gate is manually actuated, the other can be positively actuated without relying on gravity. This type of safety gate may occupy a relatively large footprint, for example, twice as high as the staging area height. It may also, especially if it comprises a three-sided, vertically translating fence, prevent access to a pallet from more than one side by personnel. It may also prevent access to a pallet from above and occupy a large fraction of the staging area footprint when in the closed position.

(27) Another type of safety gate may comprise a staging gate and a ledge gate, both of which have several articulated frame segments equipped with door rollers, such as those made by Mezzanine Safeti-Gates, Inc. Roly™. When one gate is opened, it may translate vertically while being guided along opposed tracks. The tracks may be shaped as if two garage door opening mechanisms were mounted opposite one another and the rails connected together, so opening the door on one end closes the door at the other end, and vice versa. This type of gate may have a reduced vertical envelope because the articulated gates do not need to translate their own entire vertical height to completely open. The frame to which the guide rails are mounted may occupy a large lateral footprint in both the open and closed positions, for example, at least as large as the staging area, and may also prevent lateral (three-sided) and vertical access to the staging area for pallet unloading by personnel.

(28) Another type of safety gate may comprise a staging gate and a ledge gate that comprise a swing arm with an axis of rotation parallel to the direction of pallet loading. The swing arms may form a barrier of the required height and dimensions to satisfy OH&S legislation. The swing arms may be coupled by means of, for example, torque tubes and a gearbox so that they are interlocked and only one can be open at any time. These safety gates may occupy only a small vertical envelope (for example, no higher than the staging area height) in both the loading position and

closed position, and they may allow vertical access to a pallet. If the swing arm forming the staging gate is substantially planar and only blocks one side of the staging area, fixed railing or frames may be required to block fall paths on the other two sides of the staging area, and so these safety gates may occupy a footprint at least as big as the staging area when in the closed position.

(29) Another safety gate that may be used has two sets of spring-loaded swing arms, which may be fixed to a railing and may swing inward about vertical axes to allow entry of a pallet into a staging area, for example, the PICKER PAL™ Swing Arm). Such arms may be hinged and limited so that they will not swing outwards and thus present a fall barrier. These swing arms may occupy a small platform and vertical envelope, and may not obstruct the staging area at all when in the closed position. They may not eliminate fall hazards, however, since when a pallet is in the staging area the swing arms may be held open by the pallet and thus in that moment the only barrier between personnel and the ledge is the pallet and its contents. Furthermore, such gates may prevent or obstruct withdrawal of loaded pallets from the staging area.

(30) Another safety gate system has a staging gate that is pivotally attached to a rigid frame so that it can be raised from a lowered position in which it obstructs the staging area, to a raised position, for example as disclosed in U.S. Pat. No. 2014/00047769. A pulley assembly on the frame is used to couple the staging gate to a second gate that is moveably attached to the frame, so that the gate can vertically translate to open and closed positions by being raised and lowered, respectively. The staging gate may be collapsed to a generally planar position when in the raised position. This type of safety gate may still occupy a vertical envelope twice the height of the staging area, such that the second gate may be vertically translated out of the pallet loading path.

(31) The author has discovered that previous configurations of safety gates have multiple drawbacks. One of the most common drawbacks is the safety gate having a large footprint, either vertical or horizontal. It is common to have limited space on an elevated work platform, and thus it becomes disadvantageous to have a safety gate with a large footprint as it will reduce the amount of workable space on the platform. Many safety gates also only allow access to the pallet from one side. This becomes disadvantageous if a worker is using a manual pallet jack to move pallets away from the pallet staging area. Safety gates may also cause the staging gate to rise over top of the pallet when the ledge gate is in the down position. This may prevent access to the top of the pallet, for example by an overhead crane.

(32) The author has thus discovered that several design features are advantageous in a mezzanine safety gate. It may be advantageous for a safety gate to occupy the smallest possible physical envelope, to preserve useful workspace on the elevated platform and to avoid blocking the passage of forklifts and lift trucks in adjacent aisle space. It may be advantageous to minimize the vertical envelope of a safety gate so it can be installed without restriction in many facilities. Some types of a safety gate may obstruct or limit vertical access to a pallet load, for example using an overhead crane, and/or may prevent access to the pallet from more than one side. Limiting access to one or more sides of the pallet may be disadvantageous if a worker is using a manual pallet jack to move pallets away from the pallet staging area, or in other cases. It may also be advantageous to provide a safety gate at low cost, without incidental hazards such as pinch points, and with imperviousness to corrosion which may be caused by corrosive atmospheres, such as in a chemical plant, or by frequent rinsing.

(33) Referring to FIGS. 1-10, a safety gate **10** comprises a structural frame **28**, a ledge gate **20**, and a staging gate **22**. The structural frame **28** may comprise a pair of columns **28B** spaced from one another along a peripheral edge **16** of an elevated platform **12** to define a ledge entrance **18** to a staging area **14** of the elevated platform **12**. The elevated platform **12** may, for example, be a mezzanine. The peripheral edge **16** of the platform **12** may be adjacent to a vertical wall **17**, for example the riser wall of a mezzanine in a building. The ledge gate **20** and staging gate **22** may be connected to move together relative to the pair of columns **28B**.

(34) Referring to FIGS. 1-6, the safety gate **10** may be structured to assume a loading configuration

in which the ledge gate **20** is up and the staging gate **22** is down. When moving into the loading configuration, the ledge gate **20** have been raised and pitched upwards. When moving into the loading configuration, the staging gate **22** may have been lowered and unfolded laterally away from the pair of structural columns **28B**. The raising and pitching of the ledge gate **20** may allow the ledge gate **20** to open a ledge entrance **18**. The lowering and unfolding of the staging gate **22** may allow the staging gate to enclose a staging area **14**. During use, the gates **20** and **22** may be moved as one (via mechanical linkages for example) from the unloading to the loading configuration. The staging area **14** may define a zone in which a load, such as a pallet **88**, may be placed during the transitional period between loading and unloading. The staging gate **22** and the structural frame **28** may define a staging area height **14A**, staging area depth **14B** and staging area width **14C** that define the useful envelope that is available for personnel to load and unload pallets and cargo. Referring to FIGS. **7-11**, the safety gate **10** may be structured to assume an unloading configuration in which the ledge gate **20** is down and the staging gate **22** is up. When moving into the unloading configuration, the ledge gate **20** may be lowered and pitched downward. When moving into the unloading configuration, the staging gate **22** may be raised and folded laterally against the pair of structural columns **28B**. The lowering and pitching of the ledge gate **20** may allow the ledge gate **20** to close the ledge entrance **18**. The raising and folding of the staging gate **22** may allow the staging gate **22** to expose the staging area **14**.

(35) Referring to FIG. **1**, the safety gate **10** may be installed or otherwise mounted at a suitable location and/or may be integrated within the existing safety infrastructure of the platform **12**. The gate **10** may be installed at or near or on the peripheral edge **16** of the platform **12**, to maximize area of use available rearward of the edge **16** on the platform **12**. The gate **10** may be integrated into the existing railings **48**, to allow continuity of fall barrier protection along the edge **16**, with external railings **48** running along the peripheral edge **16** of the mezzanine. The safety gate **10**, for example the columns **28B**, may connect to the external railings **48**, for example may connect to beams **48A** or columns **48B**. Referring to FIGS. **1-11**, a suitable connector or connecting mechanism, such as a guard rail connector **30**, may be used to connect the frame **28** to the railing **48**. The connector **30** may have a laterally extended mounting plate **30B**, which supports one or more U-bolt or other mount **30A**, which defines a railing beam receiving cavity that receives and grips an adjacent railing beam **48A**. Other connection mechanisms may be used, such as welding, adhesive, fasteners, brackets, and others.

(36) Referring to FIGS. **1, 3, 5** and **6**, a load lifter, such as a machine that operates a pallet jack, for further example a forklift **86**, may be used to deposit a load, such as a pallet **88**, onto the staging area **14**. The opening of the ledge gate **20**, i.e. by raising the ledge gate **20**, so it is above the staging area height **14A**, exposes the ledge entrance **18** so that the entrance **18** is unobstructed to permit a forklift **86** to deliver a piece of cargo, such as pallet **88**, into the staging area **14**. One type of load may include a pallet **88**, which is a platform structured to be lifted by a pallet jack, which carries a load **88C**, which is supported on a top **88H** of the pallet **88**. Referring to FIG. **11**, a pallet **88** may be structured to be engaged by the pallet jack, for example if the pallet **88** is equipped with tine receivers **88A** that are sized and structured to receive a set of forklift tines **86A** that may be in use inserted within the receivers **88A**, in the pallet base **88B**, to permit the pallet jack to lift, lower, and manipulate the load, for example to raise the pallet **88** to the level of the platform **12** and thereafter to advance and deposit the pallet **88** within the staging area **14**. While the forklift **86** is delivering the pallet **88**, a staging gate **22** may provide an uninterrupted railing around the staging area **14**, for example around the rear end **22A**, and the sides **22C** of the staging gate **22**, to protect the rear **14D** and sides **14E** of the staging area **14**, to keep a user out of the staging area **14** and thus eliminate a fall hazard to workers because of the exposed peripheral edge **16**.

(37) Referring to FIGS. **8** and **10**, a load lifter, for further example a forklift **86**, may be used to remove a load, such as a pallet **88**, from the staging area **14**. The opening of the staging gate **22**, i.e. by folding and raising the gate **22**, so it is above the staging area height **14A**, exposes the staging

area **14** from one or more of the rear **14D** and sides **14E** of the staging area **14**, so that the staging area **14** is unobstructed from the rear and sides to permit a forklift **86** to extract a piece of cargo, such as pallet **88**, from the staging area **14** to move the pallet **88** to another part of the platform **12**. While in the unloading configuration the ledge gate **20** is down, thus providing continuous fall protection along edge **16** to permit a user to safely access the staging area **14**. In some cases, the moving of the safety gate **10** into the unloading configuration permits a user or users to directly access, for example take apart, a load **88C** without necessarily having to use a pallet jack to move the pallet **88** out of area **14**. A staging gate **10** such as that shown in FIGS. **1-11** may be moved into an unloading configuration in which there is unobstructed access to the staging area **14**, and there is no fall hazard to personnel because of a ledge gate **20**. Cargo may be unloaded by another forklift **86**, a hand truck or manual pallet jack, an overhead crane **90** (FIG. **8**), jib crane, or other rigging equipment.

(38) Referring to FIGS. **1-11**, a ledge gate **20** may have a suitable structure. The ledge gate **20** may form a structural frame comprising one or more of beams **20D**, columns **20E**, feet **20F**, and a toe board **20C**. In some cases the gate **20** may have a planar appearance, for example if gate **20** is formed of a plate. However, a transparent or open structure as shown improves visibility and reduces material demands for manufacturing, without sacrificing safety and strength. The height and dimensions of the frame of gate **20** may be dictated by occupational health and safety law, which may, for example, prescribe a railing height of a horizontal beam **20D** of not less than 42", and/or may prescribe a toe guard. In the unloading configuration the ledge gate **20** may be the sole barrier preventing a fall hazard. Horizontal beams **20D** may define a ledge gate top end **20B**, and a base end such as a ledge gate free base end **20A**. The gate **20** may rest on the top surface of the elevated platform **12**, for instance when the safety gate **10** is in an unloading position. A free base end **20A** may be defined by toe guard or board **20C**. The feet **20F** may be equipped with a levelling or length adjustment mechanism such as a screw thread or other fastener or incremental/continuous length adjuster. A ledge gate **20** may be fabricated using suitable material, such as sheet metal and hollow structural sections and associated manufacturing methods such as bending, cutting and welding. Ground engaging feet **20F** may incorporate a levelling mechanism to compensate for slight deviations in straightness and form of a ledge gate **20**, or to adapt the gate **20** to non-level or non-planar work platforms **12**. A toe guard may be mandated by OH&S legislation to avoid personnel slipping under the lowermost horizontal beam of the gate **20**.

(39) Referring again to FIGS. **1-11**, a safety gate **10** may have a suitable structure. The staging gate **22** may comprise an end gate wall **24** that defines a rear end **22A** of the staging gate **22**. The staging gate **22** may comprise opposed side gate walls **26** that define sides **22C** of the staging gate **22**. The end gate wall **24**, opposed side gate walls **26**, and the elevated platform **12** may cooperate in the unloading configuration to define the staging area **14**. The area **14** may have a suitable shape such as a rectangular shape, for example, a shape bigger than the standard plan dimensions of at least one pallet **88**. A standard pallet size is forty eight inches by forty inches with each deck board being three and a half inches wide and five sixteenths of an inch thick, although other dimensions larger or small may be used. Clearance between the staging gate **22**, ledge gate **20**, and the sides **88E**, rear end **88F**, and front end **88G** of the pallet **88** may be at least twelve inches, although other dimensions and clearances may be used. A top **22D** of a staging gate may define the effective height of the staging gate **22**, although not in the example shown. The staging gate **22** may be made in a suitable fashion with suitable materials, for example fabricated from bolted or welded steel sections.

(40) Referring to FIGS. **1-11**, the end gate wall **24** and/or side gate walls **26** may each form a suitable respective structural frame. For example, the end gate wall **24** may comprise one or more of beams **24D**, columns **24E**, feet **24F**. Similarly, opposed side gate walls **26** of staging gate **22** may comprise one or more beams **26D**, for example whose extent may be delimited by a side gate top end **26B**. In some cases, the end gate wall **24** and/or the side gate walls **26** may include a toe board

(not shown). However, in the case of end gate wall **24** the lower most beam **24D** of the gate wall **24**, and hence the toe board if present, may act to define or limit the height **14A** of the staging area **14**. In the example shown in FIG. **1**, the toe board is not used, thus creating an upside-down U shape between beams **24D** and legs/columns **24E** and between columns **24E** and beams **26D**. In the example of FIGS. **2-11**, a toe guard such as toe board **26C** is used, to avoid personnel slipping under the lowermost horizontal beam **26D**. In some cases the end gate wall **24** may have a planar appearance, for example if gate wall **24** or side gate wall **26** is formed of a plate. However, a transparent or open structure as shown improves visibility and reduces material demands for manufacturing, without sacrificing safety and strength. The height and dimensions of the frame of end gate wall **24** or side gate wall **26** may be dictated by occupational health and safety law, which may, for example, prescribe a railing height of a horizontal beam **24D** or **26D** of not less than forty two inches. In a loading configuration, end gate wall **24** and opposed side gate walls **26** may form part of an uninterrupted barrier that mitigates a fall hazard to personnel.

(41) Referring to FIGS. **1-11**, side and end gate walls **24** and **26** may have suitable characteristics. Horizontal beams **24D** may define an end gate top end **24B**, and a base end such as an end gate free base end **24A**. The gate walls **24** or **26** may rest on the top surface of the elevated platform **12**, for instance when the safety gate **10** is in an unloading position. The feet **24F** may be equipped with a levelling or length adjustment mechanism such as a screw thread or other fastener or incremental/continuous length adjuster. An end or side gate wall **24** or **26** may be fabricated using suitable material, such as sheet metal and hollow structural sections and associated manufacturing methods such as bending, cutting and welding. Ground engaging feet **24F** may incorporate a levelling mechanism to compensate for slight deviations in straightness and form of a gate end wall **24**, or to adapt the end wall **24** to non-level or non-planar work platforms **12**. A side gate free base end **26A** may be defined by a beam **26D**, and may rest on the elevated work platform **12** when a safety gate **10** is in an unloading configuration, or it may be supported by adjacent structures such as columns **24E** of end gate wall **24**. In the example of FIG. **1**, an intermediate beam **24D** may define the base end **24A**, whereas in FIG. **2** a toe board **26C** may define the base end **24A**.

(42) Referring to FIGS. **1-11**, the structural frame **28** may have suitable components. Safety gate **10** may have a structural frame **28** that may comprise a lateral member/cross beam **28A** (such as an overhead beam, and/or in some cases a base beam). Beam **28A** may define a top **28A-1**, front side **28A-2** and rear side **28A-3**. Beam **28A** may be mounted on and bridge/span structural columns **28B**. Columns **28B** may be anchored to platform **12** by a suitable fashion, for example mounted to platform **12** via a column base plate **28C**. A column base plate **28C** may be used to mount the structural frame **28** to the elevated work platform **12**, for example by using concrete anchors to attach a base plate **28C** to a poured concrete structure. Referring to FIG. **1**, there may be provided a column footing plate **28C**, which may serve to spread the footing load over a wider area or as an endstop. In other cases, other mechanisms of mounting the frame **28** rigidly to the platform **12** may be used, and in some cases the frame **28** may rest only by gravity upon the top surface of the platform **12**. Beams in the structural frame **28**, and indeed in all frames and structural members discussed in this document may have suitable construction, such as being box beams, angle beams, I-beams, C-beams, or others.

(43) Referring to FIGS. **1-11**, ledge gate **22** may be mounted by a suitable mechanism to the frame **28**. The gate **22** may be mounted on opposed rails **36**, which may themselves be mounted to or defined by structural frame **28**, for example by one or more of fasteners, adhesives, welding, rivets, and/or brackets such as rail mounting brackets **36D**. Thus, ledge gate **22** may be mounted to move along opposed rails **36** on the pair of structural columns **28B**.

(44) Referring to FIGS. **1-11**, ledge gate **20** may be configured to move in a suitable fashion between the loading and unloading configurations. The ledge gate **20** may be oriented vertically in an unloading configuration, for example as shown in FIGS. **5-11**. The ledge gate **20** may be oriented horizontally or near-horizontally (for example within thirty degrees of horizontal) in the

loading configuration, for example as shown in FIGS. 2-4. It may be advantageous to rotate (for example pitch) the gate **20** when moving into the loading configuration because the free base end **20A** of the gate **20** may limit the maximum height **14A** of the staging area **14** through which a load can be deposited onto area **14**, so the higher the end **20A** can sit in the loading configuration (up to horizontal) the more clearance is provided. To fully open a ledge entrance **18** without rotating, a ledge gate **20** would otherwise have to translate at least as far vertically as the height **14A** of the staging area **14**, and the supporting structural frame **28** would need to be accordingly tall enough to support the gate **20** in both configurations. A safety gate **10** in which the ledge gate **20** rotates as it transitions between configurations may be smaller, lighter, and less costly.

(45) Referring to FIGS. 1-11, the rails **36** may define a suitable path of movement for the gate **20** to pitch via a pivot point or pivot points defined by the gate **20**. The path may include one or more of a straight portion **36A** and a curved portion **36B**. While moving along the curved portion **36B**, the ledge gate **22** pitches during movement. The use of curved and straight portions cooperate to partially translate and partially pitch the gate **20** during travel into the loading configuration. The curved portion **36B** may be an upper portion of the path, and the opposed rails **36** may be structured to define a path with an upside-down-J-shape. At the top end of the opposed rails **36**, the travel of the ledge gate **20** may be limited by top end stops **36C**. A ledge gate **20** may be mounted to opposed rails **36** with guide connectors **38**, for example, track rollers or door rollers. Guide connectors **38** may allow rolling motion between the ledge gate **20** and opposed rails **36** so that the gate may translate and rotate with little friction and wear, and also the guide connectors **38** may permit the gate **20** to pitch about the connectors **38**. A first set of rollers **38B** may define first pivot axis **38A** about which rollers **38B** may allow mutual rotation between a ledge gate **20** and rails **36**. A second set of rollers **38D** may define a second pivot axis **38C**, for example parallel to first pivot axis **38A**, about which second set of rollers **38D** may allow mutual rotation between a ledge gate **20** and rails **36**. The ledge gate **20** may be mounted to pitch about a pitch axis (such as axes **38A** and **38C**) perpendicular to the pair of columns **28B** and parallel with the peripheral edge **16**.

(46) Referring to FIGS. 1-11, opposed rails **36** and gate **20** may be structured such that a free base end **20A** of the ledge gate **20** pitches outward and inward relative to the columns **28B**. The free base end **20A** may pitch outward laterally away from the pair of columns **28B** and the staging area **14** when moving into the loading configuration. The free base end **20A** may pitch away from the ledge entrance **18** of the safety gate **10** as the ledge gate **20** is actuated from an unloading configuration to a loading configuration. Such pitching may increase the useful size of a staging area **14** relative to the size of a structural frame **28**, when compared with other safety gates in which a ledge gate translates vertically only or moves as a garage door without any free base end swing outside the path of movement defined by rails **36**. For example, ledge gate **20** may define a free base end **20A** and top end **20B**, and may be mounted to pitch upward and downward via first pivot axis **38A** intermediate the free base end **20A** and the top end **20B**. The ledge gate **20** may be mounted to pitch upward and downward via second pivot axis **38C** intermediate the first pivot axis **38A** and the top end **20B**. Thus, the free base end **20A** swings outward, reducing the lateral distance it has to travel toward the rear end (defined by end gate wall **24**) of gate **10** while still moving into a horizontal or near horizontal position where staging entry height is maximized. In the example shown, the ledge gate **20** may be connected to move into the unloading configuration when the free base end **20A** pitches inward laterally toward the pair of columns **28B** and the staging area **14**.

(47) Referring to FIGS. 1-11, the staging gate **22** may be structured to fold in various ways. The opposed side gate walls **26** may be pivotally attached to one or both the pair of columns **28B** and the end gate wall **24**. In some cases the gate **22** may swing in a teeter totter fashion. In other cases the gate **22** may fold in an accordion fashion. The gate **22** may be structured to assume a laterally compact configuration when in the unloading configuration, and a laterally extended configuration when deployed in the loading configuration. The gate **22** may be structured to move to lift the end gate wall **24** above a pallet **88** and load positioned in the staging area **14**, for example to lift the end

gate wall **24** over four feet in height above a rear end **88F** of the pallet **88**, and to clear a top **88D** of the pallet **88**.

(48) Referring to FIGS. **1-11**, the staging gate **22** may be structured to move via a suitable mechanical linkage (levers) to fold and unfold. Referring to FIGS. **3** and **8**, the opposed side gate walls **26** may form respective four bar mechanical linkages that fold and deploy between the unloading and loading configurations, respectively. A four-bar linkage, also called a four-bar, is the simplest movable closed-chain linkage, and may have four bodies, called bars or links, connected in a loop by four joints. Generally, the joints may be configured so the links move in parallel planes, and the assembly is called a planar four-bar linkage. Each side wall **26** may define a side gate free base end **26A**, a side gate top end **26B**, one or more horizontal beams **26D**, a toe board **26C** (not present in FIG. **1**), and feet **26F**. The side gate free base end **26A** is the portion of the side walls **26** that is nearest the floor and may comprise a toe board **26C**. The side gate top end **26B** may be defined by the horizontal beam **26D** that is the uppermost beam of the side walls **26**. The opposed side gate walls **26**, for example the uppermost beam **26D**, may be pivotally attached to the columns **24E** of the end gate wall **24**, forming a first linkage axis **32A**. The lowermost beam **26D** or toe board **26C** may be pivotally attached to columns **24E** of end gate wall **24**, forming a second linkage axis **32B**. The uppermost beam **26D** may be pivotally attached to the structural columns **28B**, forming a third linkage axis **32C**. The lowermost beam **26D** or toe board **26C** may be pivotally attached to the structured columns **28B**, forming a fourth linkage axis **32D**. There may be additional horizontal beams **26D** present, and if such is the case, then a fifth linkage axis **32F** is present on the structural columns **28B** between the third linkage axis **32C** and the fourth linkage axis **32D**, and a sixth linkage axis **32E** is present on the end gate side columns **24E** between the first linkage axis **32A** and the second linkage axis **32B**. The pivotal attachment of the side gate walls **26** may allow the safety gate **10** move between the loading and unloading configurations by folding and unfolding. When the safety gate **10** is in the loading configuration, the horizontal beams **26D** of the side walls **26** may be parallel to the elevated platform **12** of which the safety gate **10** is installed on. In order to move to the unloading configuration, force may be applied to lift the gate wall **24**, for example using an end gate beam handle **24G**, which may cause the side gate free base end **26A**, the side gate top end **26B**, the side gate toe board **26C** and any other horizontal beams **26D** present on the side gate wall **26**, to rise and fold laterally against the pair of structural columns **28B**. The end gate wall **24** may remain vertical at all times between loading and unloading configurations. Other mechanical linkages may be used.

(49) Referring to FIGS. **1-11**, the ledge gate and staging gate may be connected to move together. The ledge gate **20** may be connected to the movement of the side gate walls **26** and/or end gate wall **24**. The ledge gate **20** may move into the loading configuration when a free base end **20A** of the ledge gate **20** pitches outward laterally away from the pair of columns **28B** and the staging area **14**, while the side gate walls **26** move outward from the pair of columns **28B** and towards the staging area **14**. The ledge gate **20** may move into the unloading configuration when the free base end **20A** pitches inward laterally toward the pair of columns **28B** and the staging area **14**, while the side gate walls **26** move inward towards the pair of columns **28B** and outward from the staging area **14**.

(50) Referring to FIGS. **1-11**, as above the ledge gate **20** and the staging gate **22** may be connected to move together. Such may allow for the constant continuation of the fall protection of the external railing **48** along the peripheral edge **16** of the elevated work platform **12** despite the configuration of the safety gate **10**. When the safety gate **10** is in the loading configuration, the staging gate **22** is down, preventing access to the peripheral edge **16**. When the safety gate **10** is in the unloading position, the ledge gate **20** is down and prevents access to the peripheral edge. The constant continuation of the external railing **48** through the use of the safety gate **10** may mitigate the fall risk of being on an elevated work platform, while still allowing the loading and unloading of pallets onto the elevated work platform. The staging gate **22** and the ledge gate **20** may be lowered and raised through the use of a powered actuator, for example an electric motor or hydraulic actuator.

The staging gate **22** and the ledging gate **20** may be counterbalanced to allow them to be physically interlocked and move in opposite directions of each other through the use of a suitable mechanism such as a cable **40** and a cable guide **42**.

(51) Referring to FIGS. **1-11**, the staging gate **22** and the ledge gate **20** may be connected through a suitable connector, such as a cable **40** secured between the staging gate **22** and the ledge gate **20**. The cable **40** may be supported on a cable guide **42** mounted to the pair of columns **28B** above the ledge gate **20**. The cable **40**/connector may comprise one or more of a rope, a chain, a belt, or a strap, or other mechanism for transferring force through tension. The cable **40**, may be connected to the staging gate **22** through the use of a staging gate mount **40A**, which may be located on the side gate top end **26B** and/or end gate wall **24**. The cable **40** may extend to and be secured to the ledge gate **20**, for example through the use of a ledge gate mount **40B**. The ledge gate mount **40B** may be located at a suitable position, for example at or near a free base end **20A** of the gate **20**. The cable **40** may run through a cable guide **42** which may comprise one or more of a pulley, bushing, or a sheave. The cable guide **42** may be located above the staging area height **14A** and may otherwise be above the mounting points of the cable to the staging gate **22** and the ledge gate **20**. For example the guide **42** may be mounted on the cross beam **28A** of structural frame **28**. The cable guide **42** may ensure that the cable **40** is operated smoothly and efficiently between the staging gate **22** and the ledge gate **20**. The cable guide **42** may comprise plural guides on each side of gate **10**, for example top sheaves **42A**, located on the top wall **28A-1** of the cross beam **28A**, and the bottom sheaves **42B**, located on the front side **28A-2** of the cross beam **28A**. The location of the cable mounting points on each of gates **20** and **22** may be selected to counterbalance the movement of both gates **20** and **22** in either direction, to minimize force required to change configurations and to avoid a sudden release of energy in the form of a gate dropping rapidly in the case of an unbalanced load.

(52) Referring to FIGS. **1-11**, various mechanisms may be used to operate the safety gate **10**. End gate wall **24** may incorporate one or more of beam or column handles **24G** and **24H**, respectively. A lock **34** may be used to lock the staging gate **22** and/or ledge gate **20** in place. The lock **34** may have a bracket **34A**, a spring pin **34B**, and a pin received **34C**. The lock may have another suitable structure. A handle (not shown) may be located on the ledge gate **20**. The actuator or operation mechanism, such as any of the aforementioned handles, may be located outside the staging area **14**, to permit use only by a user outside the staging area **14**, to prevent unsafe misuse of the gate **10**. One or more pulleys or other mechanisms may be used to operate the gate **10**.

(53) Referring to FIGS. **1-11**, one or more ledge gate guards **44** may be used. For example a guard bracket **44A** may work with guard bolts **44B** and a bent guard plate **44C**. A checketer plate **46** may be used.

(54) Referring to FIGS. **12, 12A**, and **13-14**, another embodiment of a safety gate **10** is illustrated. In the example shown the cable **40** comprises a chain, such as a roller chain **40C**. The roller chain **40C** may travel along a suitable guide **42** such as sprockets **42C**. A pair of sprockets **42C** may be connected to rotate together, for example via a timing shaft **98**, to synchronize the chains **40C** on each side of the gate **10**. The sprockets **42C** may each mount for rotation to respective structural members such as posts **98B**, for further example using bearings or bushings (not shown). The posts **98B** may be mounted on the structural frame **28**. The shaft **98** may extend through the posts **98B** and may mount each sprocket **42C**. The synchronization of the rotation of the roller chains **40C** may reduce the chance of the gate **10** becoming cocked in the guides **36**. The cable **40** may comprise or be connected to a biasing device, such as a spring or springs **92**, that connect the roller chain **40C**, to one or both of the staging gate **22** and the ledge gate **20**. In the example shown, the roller chain **40C** connects at one end **40C-1** to the staging gate **22** and at another end **40C-2** to the spring **92**. The springs **92** may connect to end **40C-2** of the roller chain **40C** via a suitable connector **94A**, such as a shaft or collar. The springs **92** may connect to the ledge gate **20** via a suitable connector **94B**, such as a bracket as shown. The biasing device may comprise a stretch

limiter, such as a limit cable 96, for the biasing device. The stretch limiting cable 96 may be a non-extendible or relatively non-extendible (relative to the spring 92) cable of a predetermined length selected to restrict or prevent the overstretching of the springs 92. Referring to FIG. 12A, in the example shown, the stretch limiting cable 96 connects at one end 96A to the connector 94A and another end 96B to the connector 94B on the ledge gate 20. The stretch limiting cables 96 may prevent or restrict the springs 92 from stretching past a certain point, which may depend on the length of the stretch limiting cable 96. The springs 92 may be effective to pull the staging gate 22 tight when the gate 10 is in the unloading configuration.

(55) In the claims, the word “comprising” is used in its inclusive sense and does not exclude other elements being present. The indefinite articles “a” and “an” before a claim feature do not exclude more than one of the feature being present. Each one of the individual features described here may be used in one or more embodiments and is not, by virtue only of being described here, to be construed as essential to all embodiments as defined by the claims.

Claims

1. A safety gate comprising: a structural frame including a pair of columns spaced from one another along a peripheral edge of an elevated platform to define a ledge entrance to a staging area of the elevated platform; a ledge gate; and a staging gate; in which the ledge gate and the staging gate are connected to move together relative to the pair of columns: into a loading configuration when the ledge gate rises and pitches upward, and the staging gate moves, to allow a) the ledge gate to open the ledge entrance and b) the staging gate to enclose the staging area; and into an unloading configuration when the ledge gate lowers and pitches downward, and the staging gate moves, to allow a) the ledge gate to close the ledge entrance and b) the staging gate to expose the staging area; and in which the ledge gate is mounted to move along opposed rails on the pair of columns, the opposed rails define a curved portion of a path along which the ledge gate pitches during movement, the ledge gate defines a free base end and top end, and is mounted to pitch upward and downward via a first pivot axis intermediate the free base end and the top end.
2. The safety gate of claim 1 in which the ledge gate and the staging gate are connected to move together relative to the pair of columns: into the loading configuration when the staging gate lowers and unfolds laterally away from the pair of structural columns; and into the unloading configuration when the staging gate rises and folds laterally toward the pair of structural columns.
3. The safety gate of claim 2 in which: the curved portion is an upper portion of the path; and the opposed rails are structured to define the path with an upside-down-J-shape.
4. The safety gate of claim 3 further comprising a cable secured between the staging gate and the ledge gate, and supported on a cable guide mounted to the pair of columns above the ledge gate.
5. The safety gate of claim 4 in which: the cable is secured to the free base end of the ledge gate; the cable comprises one or more of a rope, a chain, a belt, or a strap; the cable guide comprises one or more of a pulley, sprocket, bushing, or a sheave; and the cable comprises or connects to a biasing device.
6. The safety gate of claim 5 in which the biasing device comprises a stretch limiter.
7. The safety gate of claim 1 in which: the ledge gate is mounted to pitch upward and downward via a second pivot axis intermediate the first pivot axis and the top end; and the first and second pitch axes are perpendicular to the pair of columns and parallel with the peripheral edge.
8. The safety gate of claim 1 in which the ledge gate is connected to move: into the loading configuration when the free base end of the ledge gate pitches outward laterally away from the pair of columns and the staging area; and into the unloading configuration when the free base end pitches inward laterally toward the pair of columns and the staging area.
9. The safety gate of claim 1 in which the ledge gate is oriented vertically in the unloading configuration, and horizontal, near-horizontal, or approximately forty five degrees in the loading

configuration.

10. The safety gate of claim 1 in which: the staging gate comprises an end gate wall and opposed side gate walls; and the end gate wall, opposed side gate walls, and the ledge cooperate in the loading configuration to define the staging area as having a rectangular shape.

11. The safety gate of claim 10 in which the opposed side gate walls are pivotally attached to the pair of columns and the end gate wall.

12. The safety gate of claim 11 in which the opposed side gate walls form respective four bar mechanical linkages that fold and deploy between the unloading and loading configurations, respectively.

13. The safety gate of claim 1 in which the elevated work platform is a mezzanine in a building.

14. A method comprising: moving a ledge gate and a staging gate, relative to a pair columns, into a loading configuration, the pair of columns forming a structural frame and being spaced from one another along a peripheral edge of an elevated platform to define a ledge entrance to a staging area of the elevated platform, in which the ledge gate and staging gate are connected to move together, and in which while moving into the loading configuration the ledge gate rises and pitches upward, and the staging gate moves, to allow a) the ledge gate to open the ledge entrance and b) the staging gate to enclose the staging area; moving the ledge gate and the staging gate, relative to the pair of columns, into an unloading configuration, in which while moving into the unloading configuration the ledge gate lowers and pitches downward, and the staging gate moves, to allow a) the ledge gate to close the ledge entrance and b) the staging gate to expose the staging area; and in which the ledge gate is mounted to move along opposed rails on the pair of columns, the opposed rails define a curved portion of a path along which the ledge gate pitches during movement, the ledge gate defines a free base end and top end, and is mounted to pitch upward and downward via a first pivot axis intermediate the free base end and the top end.

15. The method of claim 14 further comprising, while the ledge gate and staging gate are in the loading configuration, depositing a load into the staging area via the ledge entrance.

16. The method of claim 15 in which depositing is done using a fork lift, and the load comprises a pallet.

17. The method of claim 14 further comprising, while the ledge gate and the staging gate are in the unloading configuration, removing a load from the staging area.

18. The method of claim 17 in which: removing the load is done using a fork lift, and the load comprises a pallet; or removing the load is done using a crane.
